HEALTHCARE PATIENT READMISSION PREDICTION AND ANALYTICS USING DATABRICKS

A Mini project work submitted to the Department of Data Science,
St. Joseph's College (Autonomous), Tiruchirappalli

in partial fulfilment of the requirement for the award of the degree of

MASTER OF DATA SCIENCE

By

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Under the guidance of

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Department of Data Science
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OCTOBER 2026

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CERTIFICATE

This is to certify that the project entitled **HEALTHCARE PATIENT READMISSION PREDICTION AND ANALYTICS USING DATABRICKS** submitted to the Department of Data Science, St. Joseph's College (Autonomous), Tiruchirappalli-2, in partial fulfilment for the award of the degree of **MASTER OF DATA SCIENCE** by **HARSITHA D (REG. NO: 25PDS840)**, is a record of the original project work carried out under my guidance and supervision.

Signature of the Guide **Signature of the Head**

Submitted for the viva-voce examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**
Date Date:

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DECLARATION

I hereby declare that the mini project entitled **HEALTHCARE PATIENT READMISSION PREDICTION AND ANALYTICS USING DATABRICKS** is a record of original work done by me during the period of study, under the guidance of **Dr. K. LOURA JENCY**, Department of Data Science, St. Joseph's College, Tiruchirappalli -2.

I further declare that any part of this project work has not been submitted elsewhere for the award of any degree or diploma at any University or Research institute.

Signature

HARSITHA D

ACKNOWLEDGEMENT

I express my heartfelt gratitude to God Almighty for His abundant grace, blessings, and guidance, which provided me with the strength, wisdom, and perseverance to successfully complete this project. Without His divine support, the completion of this work would not have been possible.

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Sincerely,
Harsitha D
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ABSTRACT

Hospital readmissions are costly and are widely used as an indicator of the quality of care. Predicting which patients are likely to return, and giving administrators a clear view of readmission patterns, can support better discharge planning and resource allocation.

This project presents the development of a Healthcare Patient Readmission Prediction and Analytics system on the Databricks Lakehouse platform, using the Diabetes 130-US Hospitals dataset of 101,766 patient encounters and 50 attributes. The data is refined through a Medallion Architecture (Bronze, Silver and Gold Delta tables) with PySpark, and a 41-stage Spark ML pipeline converts it into a 2,543-dimensional feature vector for a three-class target: not readmitted (NO), readmitted after 30 days (>30) and readmitted within 30 days (<30).

Multinomial Logistic Regression and a Random Forest classifier were trained on an 80:20 split (81,225 training and 20,541 test records). Logistic Regression was selected as the final model with an accuracy of 58.51% and a weighted F1-score of 0.5327, while Random Forest reached 54.27% accuracy and a weighted F1-score of 0.3818. The four notebooks are orchestrated as a Databricks Job that completed successfully in all three recorded runs, and the results are presented in a Databricks AI/BI dashboard that reports an overall readmission rate of 46.09%.

CHAPTER 1
INTRODUCTION

1.1 Background

A hospital readmission occurs when a patient is admitted again after being discharged. For patients with diabetes, readmissions are frequent and often avoidable, and they increase treatment cost and patient risk. Hospitals therefore track readmission rates and try to identify high-risk patients before discharge.

The availability of large clinical datasets such as the Diabetes 130-US Hospitals dataset, together with scalable platforms such as Apache Spark and Databricks, has made it practical to build repeatable data pipelines and machine learning models for this task. This project leverages these technologies to develop an end-to-end readmission analytics solution deployed as an automated Databricks Job with an interactive dashboard.

1.2 Problem Statement

Manual review of patient records is slow and cannot scale to the volume of hospital data, and raw hospital data contains placeholder values, identifiers and categorical codes that cannot be used directly by machine learning models. There is a clear need for an automated and reproducible pipeline that cleans the data, builds features, predicts the readmission category of each patient, and presents the findings to hospital administrators through an accessible dashboard.

1.3 Objectives and Goals

- Ingest the raw diabetic patient dataset into Delta Lake using the Bronze, Silver and Gold layers.
- Clean the data by handling '?' placeholders, invalid categories, duplicates and identifier columns.
- Engineer machine-learning-ready features with a reproducible Spark ML pipeline.
- Train multiclass classification models (Logistic Regression and Random Forest) to predict readmission.
- Evaluate the models using accuracy, weighted precision, weighted recall, weighted F1-score and confusion matrices.
- Automate the full workflow as a Databricks Job.
- Build an interactive dashboard for executive, patient and model-performance insights.

1.4 Scope

The project covers batch processing of the historical diabetic encounter dataset (101,766 records) and a three-class readmission target. It includes data engineering, feature engineering, model training and evaluation, job orchestration and dashboarding on Databricks serverless compute. Real-time scoring, large-scale model tuning, deployment of the model as an external API, and integration with live hospital systems are outside the current scope.

CHAPTER 2
METHODOLOGY

2.1 Dataset Description

The Diabetes 130-US Hospitals dataset (diabetic_data.csv) contains hospital encounters of diabetic patients. The file was stored in a Unity Catalog volume at /Volumes/healthcare_readmission/readmission/raw_data/. It holds 101,766 encounters from 71,518 distinct patients, with no duplicate rows. The attributes cover demographics, admission details, hospital utilisation, ICD-9 diagnoses, laboratory results, 23 diabetes medication columns and the readmission outcome.

Table 2.1: Dataset Feature Description

Attribute	Details
Records	101,766 encounters
Original columns	50 (12 integer and 38 string columns in the Bronze table)
Demographics	race, gender, age (ranges such as [0-10)), weight
Admission details	admission_type_id, discharge_disposition_id, admission_source_id, payer_code, medical_specialty
Hospital utilisation	time_in_hospital, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses
Diagnoses	diag_1, diag_2, diag_3 (ICD-9 codes)
Laboratory results	max_glu_serum, A1Cresult
Medications	23 drug columns plus change and diabetesMed
Target	readmitted (NO, >30, <30)
Source	Diabetes 130-US Hospitals dataset (diabetic_data.csv)

2.2 Data Preprocessing

- **Loading:** The CSV was read with Spark using the header option and schema inference, then stored as a Delta table.
- **Missing values:** The '?' placeholders in race, weight, payer_code, medical_specialty and diag_1 to diag_3 were replaced with the category 'Unknown'. The value 'None' in max_glu_serum and A1Cresult was kept, because it means the test was not performed.
- **Invalid values:** The gender value 'Unknown/Invalid' was standardised to 'Unknown'.
- **Duplicates:** An exact-duplicate check was run; no duplicate rows were found.
- **Type conversion:** The age ranges were mapped to ordinal integers from 0 to 9.

- **Identifier removal:** encounter_id and patient_nbr were dropped to avoid identifier leakage.
- **Encoding and scaling:** Categorical columns were indexed and one-hot encoded, and numerical columns were standardised (Section 3.3).
- **Train/Test split:** An 80/20 random split with seed 42 gave 81,225 training and 20,541 test records.

2.3 System and Model Architecture

The system follows the Medallion Architecture shown in Figure 2.1: CSV file → Bronze Delta table → Silver Delta table → Gold Delta table → model training and evaluation → predictions table → dashboard. Each layer is produced by a separate notebook and stored in Unity Catalog under healthcare_readmission.readmission, so any stage can be rebuilt independently.

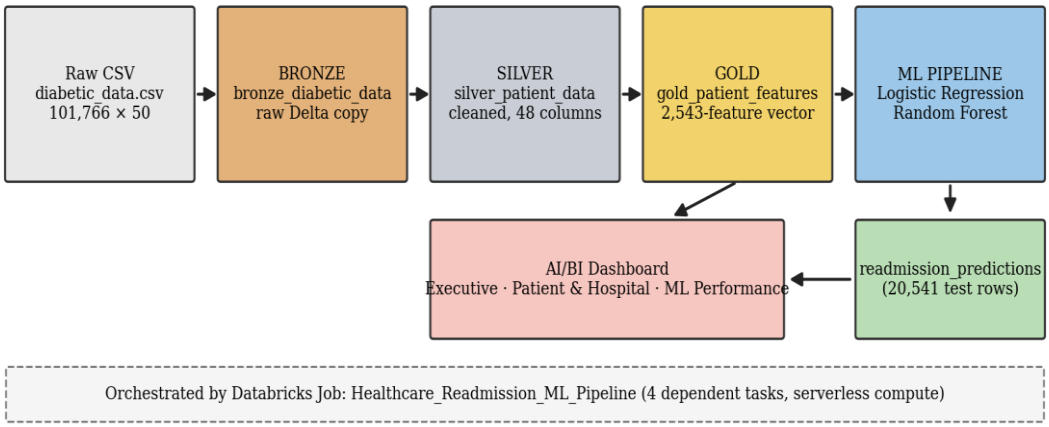


Figure 2.1: Medallion Architecture and Data Flow of the Readmission System

Two classifiers were applied to the Gold features: a multinomial Logistic Regression as a linear baseline, and a Random Forest of 30 trees as a non-linear ensemble. The model with the higher weighted F1-score is selected automatically.

2.4 Training Configuration

Both models were trained on the same training set and evaluated on the same held-out test set. Logistic Regression used the multinomial family with 100 iterations and a regularisation parameter of 0.1. Random Forest used 30 trees with a maximum depth of 6 and seed 42. Evaluation used Spark's MulticlassClassificationEvaluator.

2.5 Workflow Automation and Dashboard Deployment

The four notebooks were chained as a Databricks Job named Healthcare_Readmission_ML_Pipeline running on serverless compute, with the dependency order Bronze_Ingestion → Silver_Cleaning → Gold_Feature_Engineering → ML_Pipeline. The resulting tables feed a Databricks AI/BI dashboard titled Healthcare Patient Readmission Analytics.

2.6 Tools and Technologies

- Databricks (Free Edition) with serverless compute – platform
- Apache Spark 4.2 and PySpark – data processing
- Delta Lake and Unity Catalog – storage and governance
- Spark MLlib (StringIndexer, OneHotEncoder, VectorAssembler, StandardScaler, LogisticRegression, RandomForestClassifier) – feature engineering and modelling
- Databricks Jobs – workflow orchestration
- Databricks AI/BI Dashboards – visualisation

CHAPTER 3
IMPLEMENTATION

3.1 Bronze Layer: Data Ingestion

Notebook 01_Bronze_Ingestion reads the CSV from the Unity Catalog volume with header=true and inferSchema=true, confirms 101,766 rows and 50 columns, inspects the schema and the target distribution, and writes the data unchanged to the Delta table bronze_diabetic_data in overwrite mode. The table is then read back and its row and column counts are verified. Table 3.1 lists the tables produced by all layers.

Table 3.1: Medallion Layers and Output Tables

Layer	Notebook	Output Table	Rows	Columns
Bronze	01_Bronze_Ingestion	bronze_diabetic_data	101,766	50
Silver	02_Silver_Cleaning	silver_patient_data	101,766	48
Gold	03_Gold_Feature_Engineering	gold_patient_features	101,766	3
ML	04_ML_Pipeline	readmission_predictions	20,541	3

3.2 Silver Layer: Data Cleaning

Notebook 02_Silver_Cleaning profiles nulls and '?' placeholders for every column, checks duplicates and the target distribution, and identifies identifier and zero-variance columns. Table 3.2 shows the placeholder profile that drove the cleaning rules.

Table 3.2: Missing ('?') Value Profile

Column	Missing Count	Share of Rows
weight	98,569	96.86%
medical_specialty	49,949	49.08%
payer_code	40,256	39.56%
race	2,273	2.23%
diag_3	1,423	1.40%
diag_2	358	0.35%
diag_1	21	0.02%

The notebook then replaces '?' with 'Unknown', standardises invalid gender values, runs a duplicate removal step (0 rows removed), converts age to an ordinal integer, trims and validates the target, and drops encounter_id and patient_nbr. A final quality check confirms that no '?' values, NULLs, duplicates or identifier columns remain and that the target is valid. The result is the Silver table silver_patient_data with 101,766 rows and 48 columns.

3.3 Gold Layer: Feature Engineering

Notebook 03_Gold_Feature_Engineering builds a single 41-stage Spark ML Pipeline: 36 StringIndexers, one OneHotEncoder, a target StringIndexer, a VectorAssembler and StandardScaler for the numerical features, and a final VectorAssembler. The three integer-coded columns admission_type_id, discharge_disposition_id and admission_source_id are first cast to strings, because their numeric distances carry no meaning. A leakage review confirmed that every remaining feature is known by the time of discharge.

Table 3.3: Feature Engineering Summary

Item	Details
Excluded columns	readmitted (target); examide and citoglipton (zero variance, only 'No')
Categorical features	36, encoded with StringIndexer and OneHotEncoder
Numerical features	9: age, time_in_hospital, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses
Scaling	StandardScaler (withMean = true, withStd = true)
Label mapping	NO = 0, >30 = 1, <30 = 2
Feature vector	2,543 dimensions (sparse)

The fitted pipeline transforms the Silver data into the Gold table gold_patient_features, which keeps only the columns features, label and readmitted.

3.4 Model Training

Notebook 04_ML_Pipeline first validates the Gold table (no null or NaN labels, no null feature vectors, labels only in {0, 1, 2}) and splits it 80:20 with seed 42. It then trains the two models with the settings in Table 3.4.

Table 3.4: Model Training Hyperparameters

Parameter	Logistic Regression	Random Forest
Type	Multinomial	Tree ensemble
Iterations / trees	maxIter = 100	numTrees = 30
Regularisation / depth	regParam = 0.1	maxDepth = 6
Seed	–	42
Input	features (2,543), label	features (2,543), label
Training rows	81,225	81,225

3.5 Model Evaluation

Both models generate predictions on the 20,541 test records. Accuracy, weighted precision, weighted recall and weighted F1-score are computed with MulticlassClassificationEvaluator, and a confusion matrix is built for each model by pivoting the actual label against the predicted label. The model with the highest weighted F1-score is chosen as the best model, and its actual label, predicted label and original category are written to the Delta table readmission_predictions (20,541 rows) and verified by reading it back.

3.6 Job and Dashboard Modules

The four notebooks are registered as four dependent tasks of the Databricks Job Healthcare_Readmission_ML_Pipeline on serverless compute, so that the entire pipeline from raw CSV to predictions can be re-run with one click or on a schedule. The AI/BI dashboard has four pages: Data, Executive Overview, Patient & Hospital Analysis and ML Model Performance, with global filters for age group, gender, readmission category, admission type and discharge disposition.

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Target Distribution

Table 4.1 and Figure 4.1 present the distribution of the target. The data is imbalanced: more than half of the encounters are not readmitted, and early readmission (within 30 days) is the smallest class at 11.16%. The distribution is unchanged after cleaning and is preserved in the Gold table.

Table 4.1: Target Class Distribution

Class	Label	Meaning	Count	Share
NO	0	Not readmitted	54,864	53.91%
>30	1	Readmitted after 30 days	35,545	34.93%
<30	2	Readmitted within 30 days	11,357	11.16%

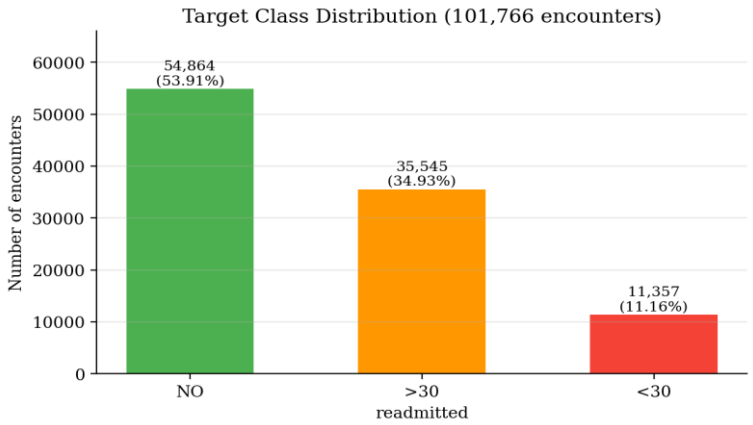


Figure 4.1: Target Class Distribution of the Readmitted Column

4.2 Data Quality Results

The Silver layer resolved every placeholder issue without losing any record: 0 duplicates were found, 0 NULLs and 0 '?' values remain, and 48 of the 50 original columns are retained. The weight column is 96.86% unknown, and medical_specialty and payer_code are 49.08% and 39.56% unknown, so these columns carry limited information even after cleaning.

4.3 Feature Engineering Output

The Gold feature vector has 2,543 dimensions for 45 input features (36 categorical and 9 numerical). The large size comes mainly from one-hot encoding the high-cardinality ICD-9 diagnosis codes (diag_1, diag_2, diag_3) and medical_specialty. All 101,766 rows and all three classes were preserved, as confirmed by the Gold verification step.

4.4 Model Performance Metrics

The two models were evaluated on the held-out test set of 20,541 records. Table 4.2 and Figure 4.2 present the results.

Table 4.2: Model Performance Metrics

Metric	Logistic Regression	Random Forest
Accuracy	0.5851	0.5427
Weighted Precision	0.5470	0.2945
Weighted Recall	0.5851	0.5427
Weighted F1-score	0.5327	0.3818

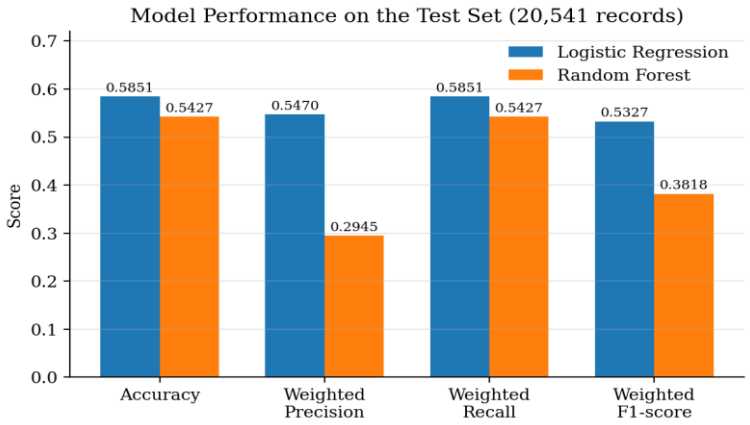


Figure 4.2: Model Performance Comparison on the Test Set

4.5 Model Comparison and Selection

Logistic Regression performed better on every metric and was selected automatically as the final model, with the highest weighted F1-score of 0.5327. The Random Forest accuracy of 54.27% is almost identical to the share of the majority class (53.91%), and its weighted precision of 0.2945 is almost exactly the square of its accuracy. This pattern indicates that the forest predicted nearly every patient as ‘NO’ and learned very little about the minority classes, which is expected for 30 shallow trees on a 2,543-dimensional sparse feature space.

4.6 Job Execution

Figure 4.3 shows the Runs view of the job Healthcare_Readmission_ML_Pipeline. Three runs were executed on 3 October, and all four tasks (Bronze_Ingestion, Silver_Cleaning, Gold_Feature_Engineering and ML_Pipeline) succeeded in every run. The total run duration increased across the runs, reaching about 20 minutes in the longest run. The job runs on serverless compute and has no job parameters.

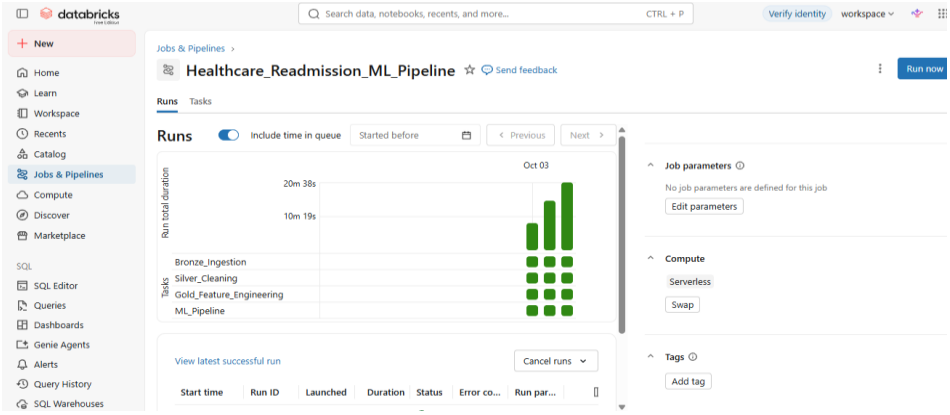


Figure 4.3: Databricks Job Run History – Healthcare_Readmission_ML_Pipeline

4.7 Dashboard

Figure 4.4 shows the Executive Overview page of the dashboard. It reports the headline figures in Table 4.3, a donut chart of the readmission class distribution and a bar chart of counts per class, with global filters on the left.

Table 4.3: Dashboard Key Performance Indicators

KPI	Value
Total patient encounters	101.77K
Readmitted after 30 days (>30)	35.55K
Readmitted within 30 days (<30)	11.36K
No readmission	54.86K
Overall readmission rate	46.09%

The overall readmission rate of 46.09% combines both readmission classes: (35.55K + 11.36K) / 101.77K. Most readmissions occur after 30 days, and early readmission is the smallest group.

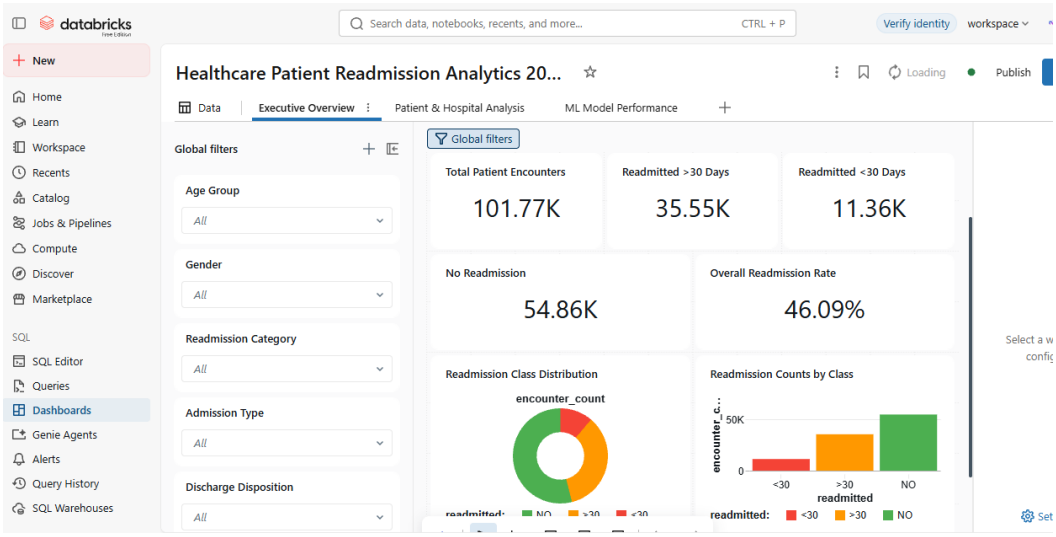


Figure 4.4: Dashboard Executive Overview – KPIs, Class Distribution and Counts by Class

4.8 Discussion

The pipeline runs end to end and its outputs are consistent at every layer: the same 101,766 rows and the same class proportions appear in the Bronze, Silver and Gold tables. Predicting readmission from information available at discharge is a hard problem. The selected model's accuracy of 58.51% is only modestly above the 53.91% obtained by always predicting 'NO', and the three-class formulation, together with the strong class imbalance, makes the task harder than the common binary version.

Several limitations explain the moderate scores. Class imbalance was not addressed, and no hyperparameter tuning or cross-validation was performed. The split is random by encounter, and because patient_nbr was removed before splitting, records of the same patient can appear in both the training and test sets. Weighted metrics are dominated by the majority class, so per-class results for the <30 class would give a more honest view of clinical usefulness. The weight column is 96.9% unknown but was retained, and the discharge disposition includes patients who died or went to hospice and therefore cannot be readmitted.

CHAPTER 5
CONCLUSION

5.1 Summary

The Healthcare Patient Readmission Prediction and Analytics system was successfully designed, developed and automated on Databricks. Raw hospital data was ingested and cleaned through Bronze and Silver Delta tables, transformed into a 2,543-feature Gold dataset with a Spark ML pipeline, used to train and compare two classifiers, and presented through an interactive dashboard. Logistic Regression was chosen as the best model with an accuracy of 58.51% and a weighted F1-score of 0.5327.

5.2 Key Findings and Outcomes

- The Medallion pipeline processed 101,766 encounters with no record loss across the layers.
- In the raw data, 96.86% of weight values, 49.08% of medical_specialty values and 39.56% of payer_code values were missing and were preserved as an 'Unknown' category.
- The target is imbalanced: NO 53.91%, >30 34.93% and <30 11.16%.
- Logistic Regression (accuracy 0.5851, weighted F1 0.5327) outperformed Random Forest (accuracy 0.5427, weighted F1 0.3818).
- The Random Forest effectively predicted only the majority class.
- The Databricks Job completed all four tasks successfully in all three recorded runs.
- The dashboard reports 46.09% of encounters as readmitted, of which 11.36K were readmitted within 30 days.

5.3 Future Scope

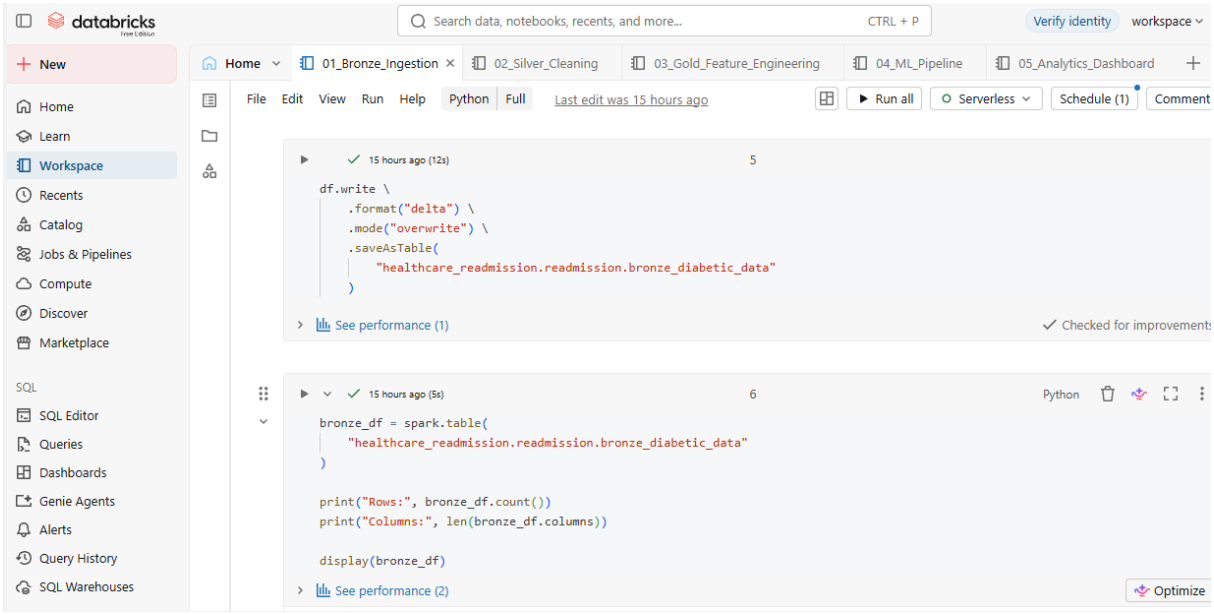
- Add a binary target (<30 days versus all others) and optimise recall and AUC-ROC for the minority class.
- Handle class imbalance with class weights or SMOTE, and try gradient-boosted trees.
- Perform cross-validated hyperparameter tuning using CrossValidator.
- Group ICD-9 codes into clinical categories to reduce the 2,543-dimensional feature space.
- Split the training and test sets by patient to remove leakage between them, and exclude expired or hospice discharges.
- Track experiments and register the final model with MLflow, and schedule the Job for regular refresh.
- Serve the model through a real-time endpoint for use at the point of discharge

BIBLIOGRAPHY

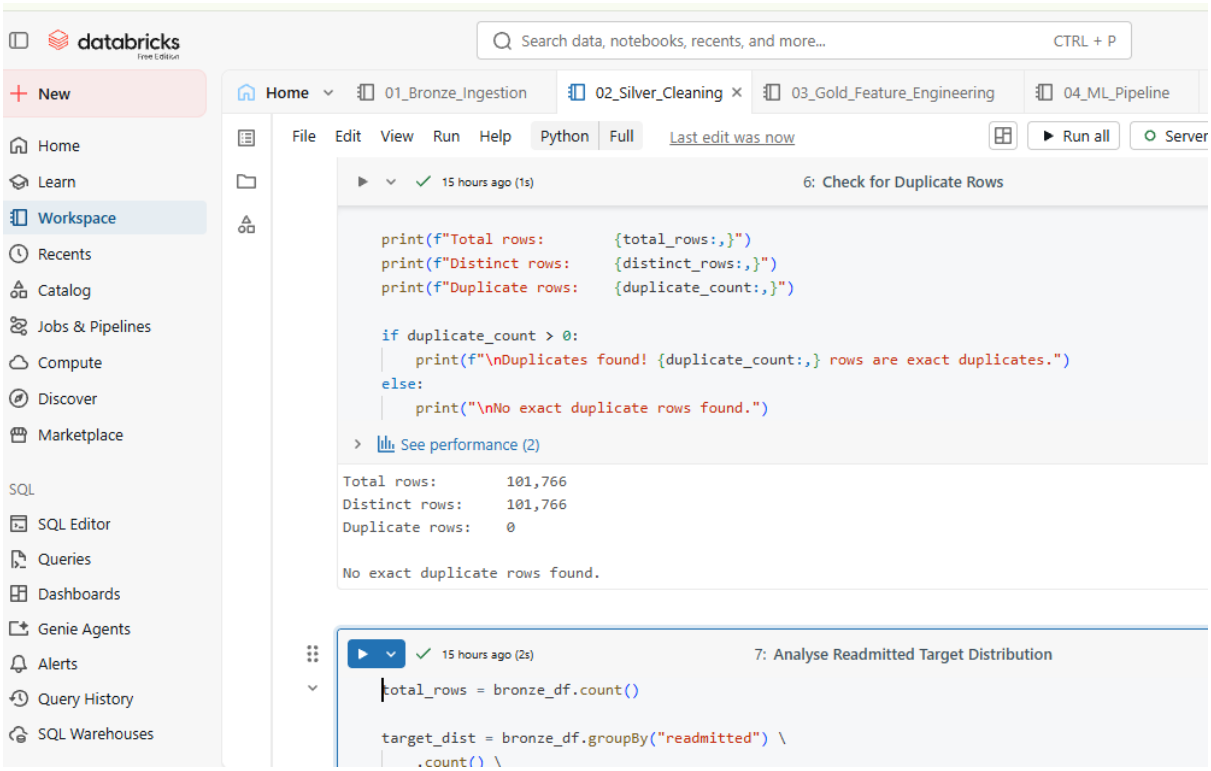
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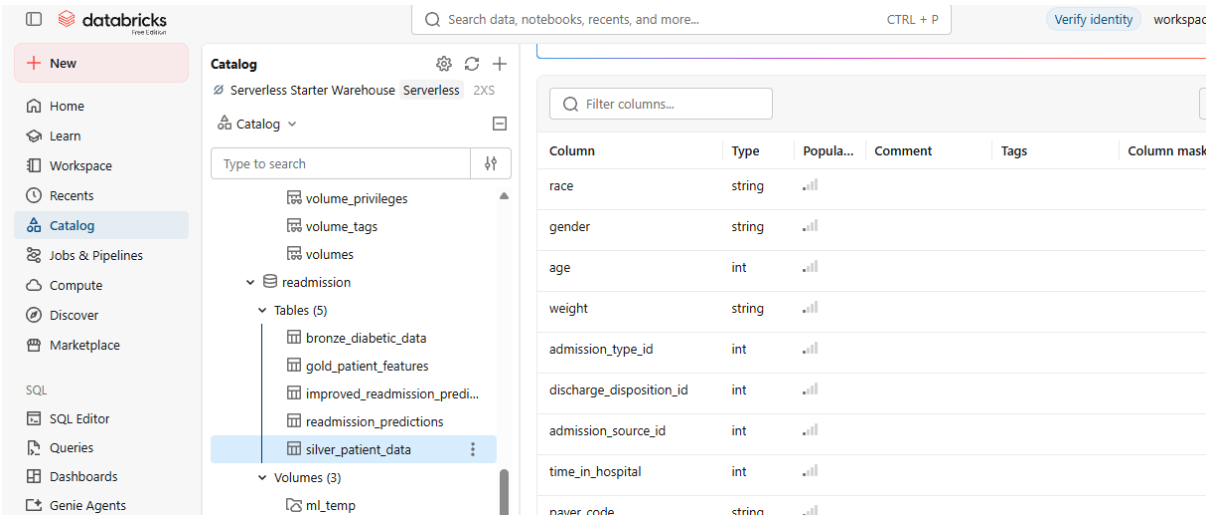
APPENDICES

Source Code – Bronze Ingestion (01_Bronze_Ingestion)

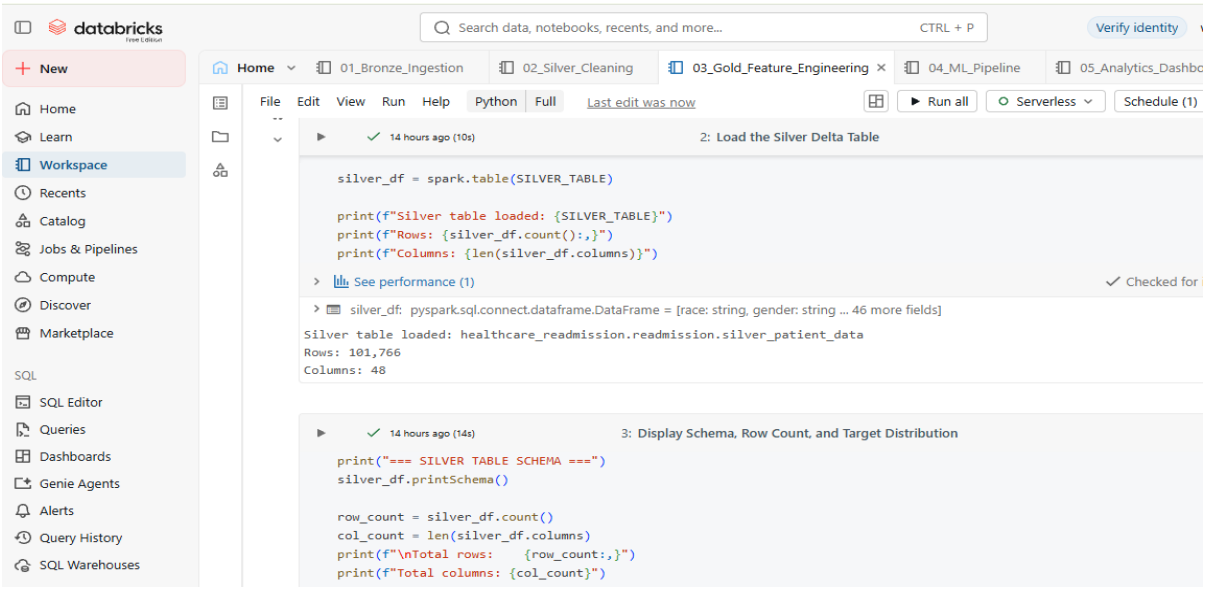


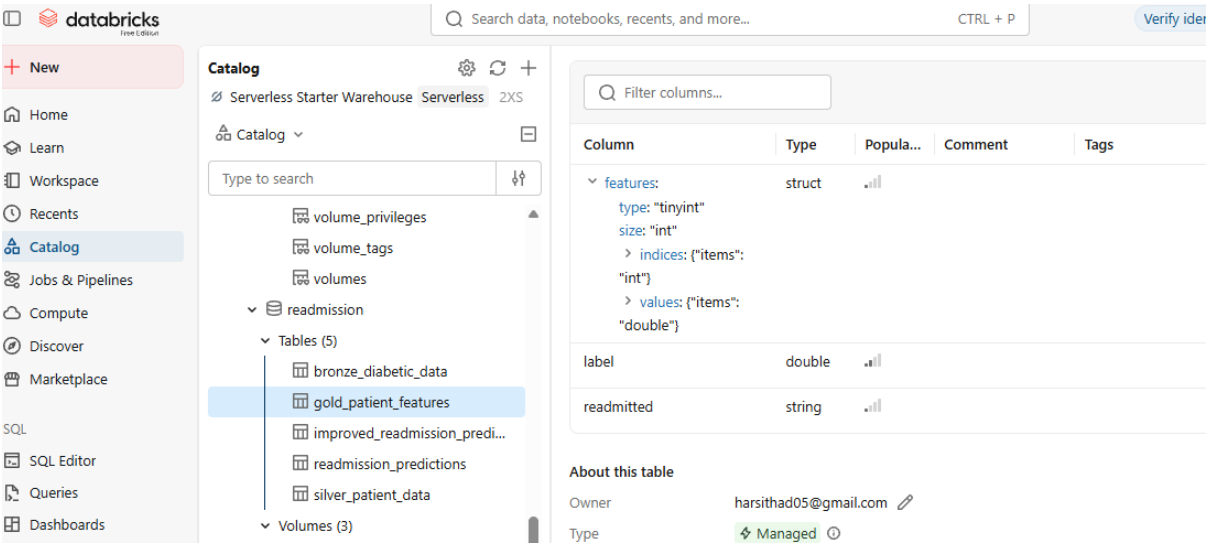
Source Code – Silver Cleaning (02_Silver_Cleaning)





Source Code – Gold Feature Engineering (03_Gold_Feature_Engineering)





Source Code – ML Pipeline (04_ML_Pipeline)

